

Review

Metal Halide Perovskites in Quantum Dot Solar Cells: Progress and Prospects

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“Go big or go home” could never be a truer statement, especially when it comes to energy. The world population is increasing, the energy use per person is growing more rapidly, and the total electricity use per person is growing even more quickly than that. To handle this demand, energy production must be increased, and it is critical for renewable sources to be used. Currently three quarters of a terawatt of power generated from photovoltaics (PVs) has been installed. At a somewhat optimistic average power-conversion efficiency of 18%, the distributed surface area of high purity optoelectronic grade semiconductor photovoltaic panels pointed at the sun is substantially larger than the area of the whole state of Rhode Island or nearly double the land mass of Luxembourg. While photovoltaic production continues to ramp to immense proportions, perovskite semiconductors are poised to greatly complement existing technology. On the other end of the size spectrum, the study of semiconductor nanocrystals or quantum dots (QDs) has led to advanced structures and deeper understanding within halide perovskite semiconductors. In this article, we show how the development of nanoscale metal halide perovskite semiconductors have gained prominence surpassing all other QD materials in terms of efficiency, and are becoming a platform for further improving technology to solve big energy challenges.

INTRODUCTION

Lead halide perovskite (LHP) semiconductors with general chemical composition of ABX_3 and crystal structure with corner-shared $[BX_6]^{4-}$ octahedra (Figure 1A) are poised to be tremendously important for low-cost solution-processed optoelectronic applications¹ due to their high optical absorption coefficient, tunable properties, long charge-carrier diffusion length,^{2–4} and most astonishingly, their ability to tolerate immense densities of structural defects with minimal electronic consequences. Before 2012, hybrid organic-inorganic materials were grossly overlooked as viable optoelectronic material contenders because of their presumed poor electron transport, but LHPs have now proven to be quite the exception and have spurred huge worldwide research efforts to explore unknown properties of “soft” hybrid (meaning crystalline “organic and inorganic”) materials. In less than a decade, the power-conversion efficiencies (PCEs) of LHP thin-film solar cells have rapidly increased from 3.6% to 25.2%⁵ as stand-alone cells, and reached 29.15% in 2-terminal tandems with silicon,⁵ making them not only a rising star in solution-processed photovoltaics (PVs), but also in many other optoelectronic applications such as optical or radiation sensing and light emission for displays, solid-state lighting, lasing, or spintronics, etc.^{6–9} Despite this rapid progress in PV PCEs, the long-term stability of the current organic-inorganic LHPs under operating conditions still requires

Context & Scale

The rise of thin-film metal halide perovskite solar cells has brought about an intense interest in their corresponding nanocrystals (NCs) or quantum dots (QDs). While these NCs retain defect tolerance similar to their bulk or thin-film counterparts resulting in high photoluminescence quantum yields, in addition, they exhibit tunable band gaps and an ability to stabilize the otherwise unstable crystal phases. Immediately upon introduction in 2015, colloidal all-inorganic lead halide perovskite NCs drew attention for lighting and display applications. The photovoltaic (PV) efficiency of $CsPbI_3$ QDs then surpassed that of all other colloidal QDs when a certified device at 13.4% was produced in 2017. Currently the record efficiency for QD solar cells has been improved to 16.6% with mixed A-site perovskite colloidal QDs. The versatility of these emerging nanomaterials in terms of ease of fabrication, band-gap tunability, and controllable surface chemistry enables various opportunities for PV such as single-junction, flexible, semi-transparent, controlled energetics with heterostructures, and multi-junction tandem solar cells that will propel this field even further.

This paper provides a comprehensive review of the development of $Cs_xFA_{1-x}PbX_3$

further research and development.^{10,11} Under ambient environmental stressors, efficient single-phase organic-inorganic LHPs (e.g., $\text{CH}_3\text{NH}_3\text{PbI}_3$) notoriously decompose into PbI_2 and volatile alkylammonium iodide (e.g., $\text{CH}_3\text{NH}_3\text{I}$ in case of $\text{CH}_3\text{NH}_3\text{PbI}_3$).¹² More complex compositions (with mixed cations and/or mixed anions) can be engineered to show dramatically better durability, but full understanding of the degradation mechanisms (such as phase segregation,^{13,14} ion migration,¹⁵ and high temperature operation) and engineering packaging¹⁶ is complex and requires a significant dedicated effort.^{17,18} On the other hand, all-inorganic LHPs (e.g., CsPbI_3), are stable in the perovskite phase only at elevated temperatures^{19,20} thus can solve some temperature related stability issues but bring about new crystal phase stability issues. These undesired challenges require additional research to find either the best composition or processing route of LHPs or engineer the entire device system toward better stability to finally push this technology forward.

Nanometer-sized colloidal perovskite quantum dots (QDs) or more broadly, nanocrystals (NCs), show high photoluminescence quantum yields (PLQYs), assisted by the absence of undesired nonradiative recombination processes. This high PLQY alone indicates that the materials are relatively benign to material defects. Additionally, LHP QDs have a tunable band gap due to the flexible compositional control that is somewhat better than in thin-film perovskites because of the benefit from crystalline strain.²² The high photoluminescence (PL) efficiency of “conventional” colloidal semiconducting QD light emitters (e.g., cadmium selenide [CdSe]) is the result of decades of extensive research on colloidal synthesis and proper surface passivation, which has culminated in their use in commercial display applications. Due to the fact that the LHP NC PL is already brighter and spectrally narrower than the best metal chalcogenide-based QDs, there exists huge potential for revolutionary optoelectronic applications.^{22–24} CsPbX_3 ($\text{X}=\text{Cl}^-$, Br^- , I^- , or mixed-halide) QDs exhibit PL tunable from deep blue to far red by adjusting the halide composition and/or QD size and shape.^{22,25,26} CsPbBr_3 QDs have been demonstrated as a highly efficient single-source photon emitter with long optical coherence times, which could be explored as building blocks in sources of indistinguishable single photons and entangled photon pairs.²⁷ Zhu et al. demonstrated that CsPbX_3 perovskite colloids (2–100 nm) can selectively photocatalyze carbon-carbon bond formation reaction, showing yet another potential application.²⁸ These all-inorganic LHP NCs also strongly absorb X-ray irradiation and exhibit tunable radioluminescence in the visible spectrum, which make them exciting candidates for scintillators.⁸ It will certainly be interesting to see what other applications may be possible with this interesting material system.

Initially, the colloidal all-inorganic LHP NCs with high photoluminescence were studied for potential lighting/display applications.²² The most suitable candidate for PV applications in this family, CsPbI_3 , suffered from structural instability upon the addition of an antisolvent, which is necessary for post-synthetic cleaning or purification of the NCs. Soon, it was found that the partly polar “magic” solvent, methyl acetate (MeOAc), enabled purification of the NCs while stabilizing CsPbI_3 in the perovskite crystal phase even after post-synthetic cleaning, and an efficient perovskite QD (PQD) PV was reported.²¹ In 2017, the PV efficiency of CsPbI_3 QDs reached a certified efficiency of 13.4%, surpassing all other colloidal QDs PV.²⁹ Currently, PQDs still hold the record efficiency for QD solar cell technology in general, with a record PCE of 16.6%.³⁰ This review highlights the unique potential of PQD solar cells, from synthesis to devices. We will discuss the current state of the art and bring forth opportunities in PQD materials research, and the related present and future pursuits in QD preparation and device architecture.

($\text{X}=\text{Cl}^-$, Br^- , I^- , or mixed-halide) QD solar cells. This is followed by a discussion of their fundamental properties, synthetic protocols, unique ion exchanges, and opportunities for advances in synthesis in terms of application for PV application. We discuss the current state-of-the-art perovskite QD PV performance and bring forth open opportunities in perovskite QD solar cell research associated with QD preparation and device architecture engineering.

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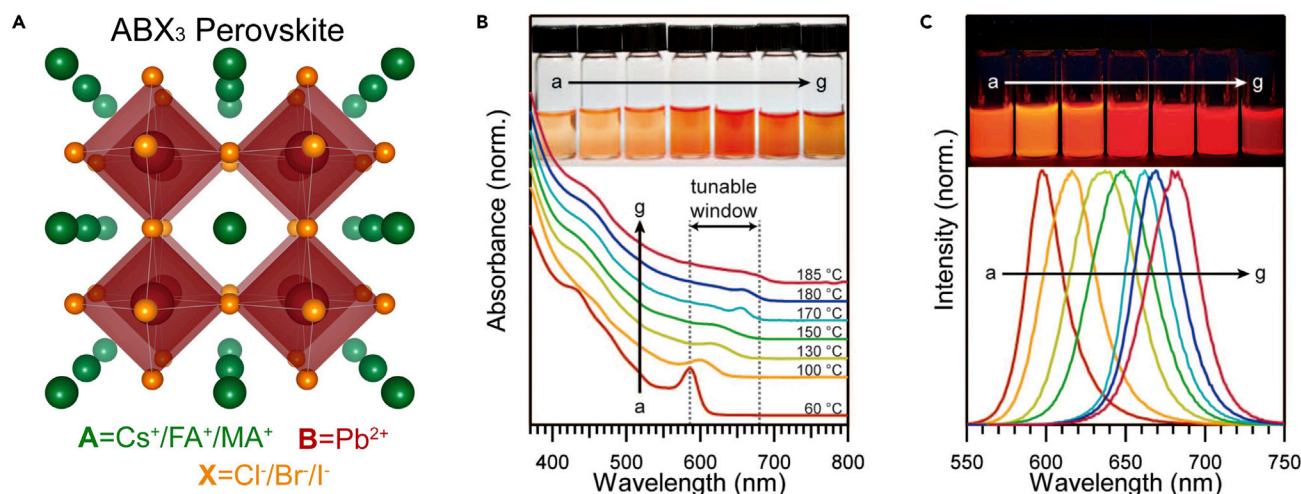


Figure 1. Material Information and Optical Properties of LHP NCs

(A) Crystal structure of APbX₃ LHP NCs.

(B) Normalized UV-visible absorption spectra of CsPbI₃ QDs synthesized at (a) 60°C (3.4 nm), (b) 100°C (4.5 nm), (c) 130°C (5 nm), (d) 150°C (6.8 nm), (e) 170°C (8 nm), (f) 180°C (9 nm), and (g) 185°C (12.5 nm), respectively. The photographs are of colloidal solutions of CsPbI₃ QDs under visible, reprinted with permission from Swarnkar et al.²¹ Copyright 2016 American Association for the Advancement of Science.

(C) Photoluminescence spectra of CsPbI₃ QDs synthesized at (1) 60°C (3.4 nm), (2) 100°C (4.5 nm), (3) 130°C (5 nm), (4) 150°C (6.8 nm), (5) 170°C (8 nm), (6) 180°C (9 nm), and (7) 185°C (12.5 nm), reprinted with permission from Swarnkar et al.²¹ Copyright 2016 American Association for the Advancement of Science.

THE EMERGENCE OF PQD SEMICONDUCTORS

Fundamental Properties of PQDs

All-inorganic LHPs have significant advantages over hybrid LHPs in terms of thermal stability and charge transport.^{31–33} However, the most suitable absorber in the family for PV applications, CsPbI₃, suffers from serious phase instability (which is discussed more in detail in the following sections). First, we focus on the reduced size three-dimensional (3D) LHPs.

Quantum Confinement Effect

In general, reducing the material crystallite size from bulk (with near infinite repeating periodicity) to a size comparable to the Bohr excitonic radius enables new and potentially advantageous properties to emerge. It allows the electronic energy levels to be engineered simply by controlling the size, in addition to the composition and/or surface chemistry.^{34–36} This ability to tune the electronic energy levels gives rise to most of the exciting properties of materials in the nanometer dimensions, as demonstrated by the now-commercialized colloidal CdSe QDs.

While the typical edge length of cubic perovskite nanocrystals is on the order of the Bohr exciton radius, the confinement properties are certainly still prominent and an intermediate confinement model has been developed to describe the size-dependent phenomena.^{37–39} The distinction between the various confinement models includes the degree to which each carrier is confined or the exciton as a whole. An example of size-dependent electronic properties in colloidal CsPbI₃ is shown in Figures 1B and 1C. The lowest energy excitonic transition of CsPbI₃ QDs occurs between 585 and 680 nm for QD sizes between 3 and 12.5 nm,²¹ respectively. Protesescu et al. reported that the exciton Bohr diameter of CsPbX₃ NCs is up to 12 nm,²² therefore exhibiting size-tunability of their band-gap energies. However, in the case of CsPbBr₃, NCs larger than 7 nm exhibit spectral signatures similar to

the bulk material. This does not imply that larger QDs do not exhibit quantum confinement effects, it only means that the confinement model applied may have weaker characteristics compared with the strong confinement model.⁴⁰ Shape and composition variation can also be used to tailor the properties, the latter being quite dominant in the field. For example, alloying between CsPbCl₃ and CsPbBr₃, and between CsPbBr₃ and CsPbI₃ enables the band gap to be tuned from deep blue to far red. PQDs provide a platform for better understanding the halide perovskite material system. Their tunable surface chemistry allows for diverse processing conditions that differ from thin-film perovskites, enabling PQDs to be compatible with many PV materials and versatile device architectures.⁴¹ In particular, QDs can be dispersed and sterically stabilized in a variety of nonpolar solvents, some of which could be more industrially friendly than those used in deposition and crystallization of thin-film LHPs. Additionally, the colloidal perovskite QD ink is already in the desired crystalline state, thus completely removing the complications of crystallization dynamics during film deposition from molecular precursors that is often observed in thin-film perovskites.

Crystal Phase Stability

The electronic structure of materials evolves from their crystal structure. Designing new materials for a particular application needs in-depth knowledge of structure-property relationships. LHPs have been extensively studied in terms of their crystal structure and related electronic properties.^{42–44} A majority of “all-inorganic” bulk CsPbX₃ materials are thermodynamically stable in the non-perovskite orthorhombic (δ) phase ($Pnma$) with edge-sharing [PbI₆]⁴⁻ octahedra in ambient conditions. The large band gap (>2.5 eV) and poor electrical properties make the non-perovskite phase material an inferior absorber for PV applications.³¹ In contrast, the corresponding colloidal nanocrystals of CsPbX₃, most importantly the material of interest in PV applications, i.e., CsPbI₃ QDs, due to high surface area and surface strain, are stable in the perovskite phase and can be stored for months in ambient conditions.⁴⁵ Bulk CsPbI₃ has strongly temperature-dependent crystallographic properties,⁴⁴ and can exist in different crystal phases at different temperatures. Even the black perovskite phase can have different space groups depending on the nature and amount of tilting of the corner-shared [PbX₆]⁴⁻ octahedra: α -phase with undistorted octahedra (with Pb-I-Pb angle of 180°) stabilizes above 300°C, whereas β - and γ -phases with distorted octahedra (Pb-I-Pb angle smaller than 180°) stabilize at less than 300°C. These small changes in crystallographic parameters not only affect the electronic properties, but also act as decisive factors in determining the amount of polarity or ferroelectricity in the material, which in turn affect the device parameters like hysteresis or charge separation. It is to be noted here that the concept of ferroelectric PV has been known particularly in oxide perovskites for a while.^{46,47} The complexity of their crystal structure and instability of the perovskite phase in ambient conditions make it difficult to study their crystallographic properties experimentally. However, PQDs can be used as a model system; they can be stabilized in the useful perovskite phase at ambient temperatures. However, it should be noted that one limitation of small crystallite sizes is that it becomes difficult to extract the experimental crystallographic information due to Scherrer broadening.

Initially, CsPbX₃ PQDs were reported to crystallize in the cubic $Pm\overline{3}m$ perovskite phase (with undistorted [PbX₆]⁴⁻ octahedra).²² However, later studies using synchrotron-based X-ray diffraction confirmed the presence of octahedral distortion in CsPbBr₃ and CsPb(I_xBr_{1-x})₃ QDs.^{48,49} Crystallographic studies on CsPbI₃ PQDs were recently reported by Zhao et al., which explores how the size-dependent tensile surface strain results in crystallographic tilting.³⁷ There are plenty of additional

opportunities to explore their structural evolution or changes as a function of size, surface chemistry, strain, or temperature or pressure, etc. The same is true for colloidal FAPbI_3 and $\text{FA}_{1-x}\text{Cs}_x\text{PbI}_3$ PQDs, where all compositions can be stabilized in the black perovskite phase at ambient conditions. It is to be noted here that along with CsPbI_3 , colloidal FAPbI_3 and $\text{FA}_{1-x}\text{Cs}_x\text{PbI}_3$ PQDs are also actively explored in solar cell applications.^{45,50–52} Full range tuning of A-site mixing using Cs and FA is challenging and has not been reported for bulk or thin-film perovskites, whereas the particular composition $\text{Cs}_{0.5}\text{FA}_{0.5}\text{PbI}_3$ has recently shown to have the certified record efficiency of 16.6% for any QD solar cell.³⁰ One may argue from a geometrical point of view that any composition of $\text{Cs}_x\text{FA}_{1-x}\text{PbX}_3$ should be stable in a perovskite phase with corner-shared octahedra (Goldschmidt tolerance factor, GTF, between 0.8 and 1), but the GTF is only a guideline and does not always correctly predict energetically favored crystal phases.

Defect Tolerance

In general, semiconducting QDs are prone to a large variety of defects. Although NC crystallization enables many crystal defects to be expelled to the surface during growth, the high surface area of NCs makes it more likely that surface defects can arise and limit the PLQY. For example, removing the surface Cd^{2+} ions from CdSe QDs leads to non-bonding or weakly bonding localized orbitals, which lie in the mid-band-gap region. These orbitals act as trap states, which facilitate nonradiative recombination to reduce the overall PLQY. However, in PQDs the energetic position of most trap states is in the conduction and valence bands, rather than within the band gap, leading to negligible effects on the PLQY.^{53,54} This is why, unlike the conventional metal chalcogenide or pnictide QDs, the as-synthesized LHP QDs can exhibit ~90% PLQY and are actively researched for LEDs, and do not necessarily need shelling with wider band-gap semiconductors to enhance their PLQY.^{55,56} Despite the high defect tolerance, LHP QDs are certainly not entirely defect-free, as PLQYs are normally reported in the range of 80%–95% in the green-red region, and down to 10%–20% at the blue border region in case of CsPbX_3 .²⁶ Vacancies primarily in the A- and X-sites are commonly observed due to the sufficient low formation energy^{57,58} (for instance, <0.5 eV in LHPs as compared with ~2–3 eV in oxide perovskites like CaTiO_3) and these vacancies lead to shallow defect states due to the special ionic bonding in LHP QDs. Nonetheless, the defect tolerance and PLQY values of LHP QDs are higher relative to other non-passivated chalcogenide QDs, and after further exploring post-synthetic ligand treatments and doping methods, the PLQY of LHP QDs has been reported near 100%,^{59,60} even for those emitting in the blue-wavelength region.^{61,62}

Synthesis of CsPbX_3 and FAPbX_3 Nanocrystals

As shown in Figure 1A, LHPs with general chemical formula of ABX_3 , where A is cesium (Cs^+), formamidinium (FA^+ , $\text{CH}(\text{NH}_2)_2^+$), or methylammonium (MA^+ , CH_3NH_3^+), B is Pb^{2+} and X is single or mixed halides (Cl^- , Br^- , or I^-), have corner-shared $[\text{BX}_6]^{4-}$ octahedra in their unit cell. As compared to the traditional metal chalcogenides and pnictides, the chemical bonding in ternary LHPs is intrinsically much more ionic.^{22,53} As a result, LHP-based QDs (or NCs) can form within seconds during the colloidal hot-injection synthesis, even near room temperature, using various precursors (Figure 2A).^{22,26} In general, as will be discussed below, despite being forgiving materials, the synthesis still leaves plenty of room for improvement with respect to better monodispersity, shape control, composition control, and surface properties.

The first colloidal synthesis of LHP NCs was reported back in 2014 by Schmidt et al., where 6 nm-sized MAPbBr_3 nanoparticles, stabilized by a long or medium chain alkyl

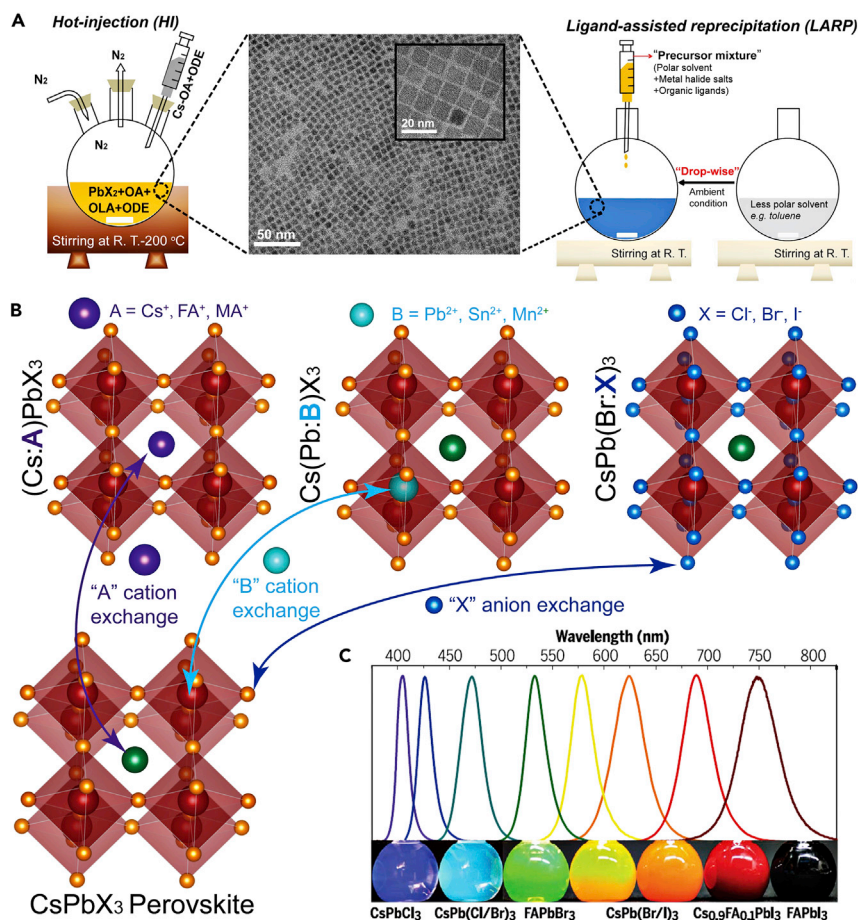


Figure 2. Synthesis and Ion Exchange of PQDs

(A) Schematic illustration of classical hot-injection and ligand-assisted reprecipitation method for the synthesis of colloidal perovskite nanocrystals (oleic acid [OA], oleylamine [OLA], octadecene [ODE], PbX_2 [$\text{X}=\text{Cl}, \text{Br}$ or I]), transmission electron microscopy (TEM), and high-resolution TEM images of CsPbI_3 QDs synthesized at 180°C.

(B) Schematic showing "A" cation, "B" cation, and "X" anion exchange of the LHPs enables band-gap tuning over the whole visible spectrum (as shown in insert C), as well as the thermodynamic stability tuning.

(C) Photoluminescence spectra and photographs of colloidal perovskite QD solutions showing exciton emission tuning over the whole visible spectrum aided by "A" and "X" site exchanges, reprinted with permission from Kovalenko et al.⁵⁷ Copyright 2017 American Association for the Advancement of Science.

ammonium bromide were synthesized by reacting with MABr and PbBr_2 in 1-octadecene and oleic acid.⁶³ This pioneering report made it possible to prepare homogeneous LHP NC thin films via spin coating of these nanoparticle solutions. A year later, Zhang et al. first demonstrated a "ligand-assisted reprecipitation" (LARP) method for synthesis of colloidal MAPbX_3 nanocrystals at room temperature,⁶⁴ where the nucleation and growth of the particles took place in a less-polar solvent, like toluene, upon addition of the precursor mixture of MAX, PbX_2 , oleylamine, and oleic acid in a polar solvent (N-dimethylformamide [DMF]) (as illustrated in Figure 2A). The transformation of these precursors into crystalline colloidal nanoparticles was induced by the solubility difference before and after solvent mixing. LARP was widely adopted for a variety of colloidal perovskite nanocrystals synthesis,^{65–67} but it was the hot-injection method reported by Protesescu et al. in 2015²² (shown in Figure 2A) that

really ignited the research efforts of LHP QDs, particularly all-inorganic (CsPbX_3) NCs.

In terms of size control that leads to highly monodisperse particles, LHPs are far behind the metal chalcogenide NCs. Unlike metal chalcogenide NCs, where fast nucleation is followed by slow growth which eventually leads to highly monodispersed particles, the nucleation and growth occur rapidly for LHP NCs, as expected for an ionic metathesis reaction.^{22,68} During hot-injection colloidal synthesis of metal chalcogenide NCs, upon mixing of precursors, a molecular complex is formed first, which leads to the monomer addition for slow growth of NCs⁶⁸ ranging in size from 2 nm to about 10 nm, with typical size distributions of around 3%–5%.^{69–71} However, in the synthesis of LHP NCs, no such molecular complex or monomers are known. Thus, the nucleation step and the growth stage cannot easily be controlled kinetically, which leads to some control over general size but with much larger size distribution. The mean size of CsPbX_3 NCs can be tuned in the range of 4–15 nm by reaction temperature rather than growth time, however, the controllability and uniformity of the QD size are still insufficient.²² Nonetheless, there have been noteworthy synthetic improvements. Rather than by *kinetic* means, Dong et al. demonstrated *thermodynamic* control over the synthesis of CsPbX_3 NCs by adding excess metal halide salts to the synthesis mixture to obtain precise control over size in sub-10 nm regime, while retaining narrow size distribution in CsPbBr_3 NCs.⁷² The size control was achieved by varying the reaction temperature and the Br:Pb ratio. Since the X^- in CsPbX_3 NCs is mobile and highly labile, by adding excess metal halide salts to the reaction system, the nucleation and growth kinetics are controlled to obtain size control and uniformity. This synthesis results in two-dimensional (2D) nanoplatelets along with the 3D perovskite nanocubes, which can be separated post-synthetically for shape-pure perovskite colloidal materials. Various other metal halides, such as GeI_2 have been used to improve material properties and deserves more attention to fully understand mechanistically.⁷³ Imran et al. demonstrated the synthesis of nearly monodispersed (6.3 ± 0.9 , 9.4 ± 0.7 , 12.9 ± 1.0 , and 18.7 ± 1.2 nm), shape- and phase-pure CsPbBr_3 nanocubes with larger sizes (7–18 nm) using a variety of secondary amines (e.g., didodecylamine) instead of the commonly used primary amine (oleylamine) as the capping ligand,⁷⁴ which was ascribed to the inability of secondary amines to find the right steric conditions at the nanocrystals surface, consequently restraining the formation of low-dimensional structures. Interestingly, the CsPbBr_3 nanocubes with sizes above 10 nm also exhibit a sharp excitonic feature in the absorption spectrum, despite being larger than its exciton Bohr radius (7 nm).²² Synthesizing highly monodispersed colloidal nanocrystals has tremendous implications for optoelectronics. Long-range self assembly of NCs could enhance interparticle electronic coupling and carrier transport and is only possible with tight control of the size distribution.⁴⁰ It has been recently shown that such superlattice structure from self-assembled CsPbX_3 nanocrystals enables “superfluorescence,” which was not previously observed in any kind of colloidal nanomaterials.⁷⁵

Due to their unique defect-tolerance characteristics, the PLQY for as-synthesized CsPbBr_3 nanocrystals has been reported to be greater than 90% without any further treatment or shelling or encapsulation.⁷⁶ Unfortunately, a hidden caveat is that LHP nanocrystals often show reduced PLQY after the antisolvent purification process,^{59,68,77} mainly due to loss of the labile surface oleylammonium halide ligands or due to loss of the outer $[\text{PbX}_2]$ shell.⁷⁸ Synthetic strategies and post-synthetic treatments to overcome this are being reported to improve PLQY. Nenon et al. showed surface halide vacancies as the predominant source for charge-carrier trapping and reduced PLQY, and demonstrated that using softer Lewis bases (e.g.,

alkylphosphonates, fluorinated carboxylates, and sulfonates) as passivating ligands for the under-coordinated Pb (relatively soft Lewis acid) significantly improved the PLQY.⁷⁹ Koscher et al. and DiStasio et al. reported that a post-synthetic treatment with thiocyanate salts⁵⁹ and lead bromide⁸⁰ improves the PLQY to near unity in CsPbBr₃ nanocrystal colloidal solutions and solid-state films, respectively. Other various strategies to increase the PLQY of the perovskite nanocrystals include designing better passivating ligands like: (1) bidentate ligands (2,2'-iminodibenzoic acid),⁸¹ resulting in stronger binding to the perovskite surface and hence reduced surface trap states; (2) Zwitterionic molecules (3-(N,N-dimethyloctadecylammonio) propanesulfonate)⁸² for kinetically stabilizing NCs after isolation and purification due to strong adhesion to the NC surface by the chelation effect; (3) 3-aminopropyltriethoxysilane (APTES)⁸³ as anchoring on the NCs' surface via hydrogen bonds with its terminal amine (ammonium) group, while simultaneously allowing for the polymerization of silane primers to a polymeric network surrounding the NCs' surface; and (4) di-dodecyl dimethyl ammonium bromide (DDAB)⁸⁴ for the coordination of the Br⁻ anions with the surface charged metal centers (Cs⁺ or Pb²⁺), with the DDA⁺ attached on QD surface. Meanwhile, direct synthesis using trioctylphosphine (TOP)-PbI₂ precursor complex has shown near-unity PLQY in case of CsPbI₃ nanocrystals.⁶⁰ TOP is reported to coordinate to the surface of the resulting QDs due to the presence of lone pair electrons on the phosphorus atom, supplying a more complete surface passivation for the QDs with negligible charge-trapping pathways. Using conventional long-chain alkylammonium halide precursors and performing the synthesis at unusually high temperatures of 220°C–260°C results in near 100% PLQY in all the CsPbX₃ nanocrystals,⁸⁵ where the desired amount of preformed alkylammonium halide has a dual purpose of providing halide ions and the capping ammonium ions. All of these synthetic and post-synthetic efforts open the door to rational design of trap-free CsPbX₃ nanocrystals for light emission applications, based on LHP materials.

The hybrid perovskite QDs with an organic A-site are less explored, as compared to the all-inorganic CsPbX₃ nanocrystals. In particular, FAPbI₃ is of interest to the solar cell community because of its slightly narrower band gap compared with MAPbI₃ and CsPbI₃, and also because of its longer PL lifetime (~70 ns)⁷ and slowed hot carrier cooling.⁸⁶ As for the other optoelectronics properties, colloidal FAPbI₃ nanocrystals show amplified spontaneous emissions (ASE) in the near infrared region.⁷ From the crystal tolerance factor point of view, one would expect improved phase stability when the A-site in APbX₃ has a mixture of Cs and FA as compared with pure Cs, and in fact has been shown to be true experimentally.⁷ These improved properties inspire additional research into FAPbI₃ for optoelectronic applications. The FAPbX₃ colloidal nanocrystals can be synthesized either by LARP method⁶⁵ at room temperature or hot-injection method at <100°C.⁷ Using the hot-injection method, various compositions of Cs_xFA_{1-x}PbI₃⁷ or Cs_xFA_{1-x}Pb(Br_{1-y}I_y)₃⁸⁷ can also be synthesized. Compositions that cannot be accessed via direct synthesis can be realized using post-synthetic transformations as discussed in the following section.^{50,52} It is likely that knowledge of CsPbX₃ surface chemistry can be extended to propose design principles for the preparation of monodispersed, shape- and phase-pure and trap-free FAPbX₃ nanocrystals.

Ion Exchanges

The more ionic nature of LHP QDs is beneficial for the post-synthetic management of chemical compositions.^{88,89} Post-synthetic ion exchanges preserve the overall QD size and shape, allowing for compositionally tunable materials.^{26,90,91} Anion exchange in CsPbX₃ QDs was first reported by Nedelcu et al.²⁶ Halide exchange takes

place rapidly at room temperature, simply by mixing two QD solutions of different compositions owing to the small activation energies for defect or ion migration inside the nanocrystals and the high dynamic nature of the surface ions.⁹² Ion exchange has been further demonstrated for one-dimensional (1D) and 2D perovskite NCs,⁹¹ which creates more avenues for composition and property tuning. The full compositional tuning in LHP QDs is more versatile than ion exchange in conventional metal chalcogenide or pnictide NCs, where cation exchange is easier to control than anion exchange.²⁴ In LHP QDs, the A-site, B-site cations and X anion can be partially or fully exchanged with other similar but foreign ions (for instance, Cs^+ with Rb^+ , FA^+ , or MA^+ , Pb^{2+} with Sn^{2+} , Cd^{2+} , and Zn^{2+} , and Br^- with Cl^- , Br^- with I^- , and vice versa). Additionally, cation- or anion exchange in LHP QDs can be tuned arbitrarily over the full compositional range, even if some parts of this compositional space may be more difficult to achieve in thin-film LHPs, as demonstrated in the case of $\text{Cs}_x\text{FA}_{1-x}\text{PbI}_3$ alloys.^{50,52} By manipulating A- or B-site cations via post-synthetic transformations, new perovskite NC materials which cannot be directly synthesized may be obtained, and the NC's ambient stability can potentially be improved.⁵⁰ Moreover, complex compositions in a wider range with simultaneous A- and X-site alloying can be realized in a single post-synthetic step.⁵²

The simplest post-synthetic alloying of perovskite NCs can be achieved in two different ways: either by mixing two different compositions of individually synthesized nanocrystals (for example combining CsPbCl_3 with CsPbBr_3 nanocrystals) or by exposing nanocrystals to salts of A, B, or X ions (for example exposing CsPbCl_3 to oleylammonium bromide).^{24,89,93} Other examples of post-synthetic transformations include photoinduced anion exchange of CsPbX_3 , where these nanocrystals are photoexcited in a solution of dihalomethane.⁹⁴ Both the X-site anion exchange and the A-site alloying enable a wide range of band-gap tuning and can provide better phase stability in bulk LHP films.^{30,50,95} However, in comparison with singly charged highly mobile halide ions, the rigid cationic sublattice increases the difficulty of such cation-exchange transformations either in solution or solid state. Initial attempts for such cation-exchange transformations led to decomposition of the parent nanocrystals.²⁶ Nonetheless, post-synthetic A-site alloying has been explored and developed in the last couple of years, examples being full-range alloying of the A-site by mixing CsPbI_3 and FAPbI_3 colloidal nanocrystals,^{50,95} and the complete transformation of MAPbX_3 into FAPbX_3 by a solid-liquid-solid cation exchange.⁹⁶ The successful full-range alloying of the A-site may be attributed to the small activation energy (~ 0.65 eV) for the miscibility between Cs^+ and FA^+ cations; however, it is higher than the value reported for X-site exchange in bulk LHP films. B-site Sn^{2+} exchange with Pb^{2+} at controllable levels have been demonstrated in hybrid organic-inorganic bulk perovskite thin films.⁹⁷ However, there are still no successful reports of full-range B-site exchange in perovskite NCs, and it has been suggested that such an exchange is facilitated by X-site vacancy migration.⁹⁰ Additional research of A- and B-site exchange could enable efficient and lead-free LHP NCs.

Opportunities for Advances in Synthesis

Improved Size Control

For PV applications, NC synthesis has many important implications. For one, the colloidal synthesis is an extra step not required for polycrystalline perovskites. Thus, while the synthesis enables well-controlled surfaces, eliminates defects, and does not rely on the same crystallization process governed by the substrate, it would likely add expense to a final module.⁹⁸ Ultimately the goal would be for the synthesis to be performed at the lowest cost possible, with efficient use of precursors, using low-cost (recyclable) solvents, and would yield nanocrystals with tight size

distribution. Since the size of PQDs used in PV applications are generally larger than the Bohr diameter,³⁷ the energetic disorder due to size dispersion is minimized compared to other highly confined QDs, such as PbS, PbSe, and CdSe.^{99,100} However, narrower size distribution could potentially improve efficiency of QD solar cells in more than one way. First, electron transport could be improved within smaller QDs as larger QDs act as a band tail¹⁰¹ or a shallow trap that impede transport.¹⁰² Second, the open-circuit voltage (V_{OC}) of QD solar cells can be limited by the smallest band gap (largest size) of QDs near the contacts.¹⁰² Improving QD homogeneity and size uniformity will improve PV efficiency by minimizing such losses. While controlled experiments like these have not yet been reported, it is feasible that a more controlled synthesis would benefit PQD solar cells.

Heterostructures

Core/shell nanocrystals have demonstrated benefits in other QD PV architectures for reduced exciton recombination at core surface sites,¹⁰³ enhanced charge transport and improved the material stability.¹⁰⁴ However, well-controlled core/shell perovskite nanocrystals are still in their infancy. Zhang et al. first reported core/shell structure by capping CsPbI₃ NCs with PbS.¹⁰⁵ The thin PbS shell passivation significantly enhanced the PL efficiency, reduced the Stokes shift, narrowed the PL bandwidth and increased the stability of CsPbI₃ NCs. Ning et al. and Liu et al. demonstrated that perovskites and PbS can exhibit coherence in their lattice fringes due to their minimal lattice mismatch (<5%),^{106,107} pointing out a great possibility in obtaining high-quality perovskite/PbS nanostructures. In addition to core/shell heterostructures, other linear heterostructures or Janus particles could offer new physics for a variety of PV applications. Zhang et al. reported the preparation of Cd/Pb Janus heterostructured particles¹⁰⁸ and demonstrated that multiple exciton generation (MEG) in Janus-like hetero nanostructures is enhanced over that of single-component and core/shell nanocrystal architectures.¹⁰⁹ The perovskite Janus particles have recently emerged to overcome the poor stability of CsPbX₃ nanocrystals. Hu et al. reported monodisperse and highly stable CsPbX₃/SiO₂ Janus nanoparticles through an interfacial synthesis method.¹¹⁰ Indeed, these novel nanostructures provide directions to further improve QD PV performance, however, the current status is still far from direct implementation in solar cell devices. Well-defined device architectures need to be explored to utilize the advantages of these functional nanomaterials, but nonetheless it still presents an open opportunity.

PQD SOLAR CELLS

LHP QDs offer opportunities to explore a variety of concepts not easily achieved in thin-film solar cells, such as quantum confinement or/and size effect-induced tunability of electronic states,¹¹¹ and potentially larger device structure tunability and use in multijunctions. PV devices based on solution-processed QDs were first investigated around 2000.^{112,113} Since then, great success has been achieved in colloidal lead chalcogenides (PbS or PbSe) QDs-based PV devices^{41,114–118} with the certified PCE exceeding 12.0%,¹¹⁷ dominating early QD PV research. The progress was further amplified when CsPbI₃ QD arrays were first used in PVs in 2016, where a PCE of 10.77% was achieved.²¹ So far, the state-of-the-art PQDs solar cells have been certified by Newport Corporation with stabilized power output efficiency (SPO) of 16.6%.^{5,30} This progress of PQD solar cells is summarized in Figure 3A. Despite the high surface area of QDs, CsPbI₃ PQD solar cells show exceptionally high V_{OC} up to 1.27 V,¹¹⁹ which is distinctly larger than that in CsPbI₃ polycrystalline thin-film solar cells and is on par with the best of all perovskites with comparable band-gap values. Enhanced V_{OC} values have also been witnessed in CsPbBr₃¹²⁰

and $(\text{FA}_{1-x}\text{Cs}_x)\text{Pb}(\text{I}_{1-x}\text{Br}_x)_3$ ⁵² PQD solar cells. We discuss these details further in the following section.

Progress of PQD Solar Cells

As shown in Figure 3B, the first PQD solar cell was demonstrated in 2016,²¹ immediately after the first reported hot-injection synthesis of CsPbX_3 perovskite NCs.²² Understanding the surface properties of CsPbX_3 QDs is critical for high-efficiency PQD solar cells. Surface ligand properties control the dot-to-dot carrier transport necessary for the desired performance of solar cells. In order to prepare as-synthesized QDs for fabrication into a solar cell, the QDs must be washed to remove excess unreacted precursors. Methyl acetate (MeOAc) was first demonstrated as an antisolvent that can precipitate the QDs without inducing agglomeration or a phase transition during the QD purification process (Figure 3C). After the MeOAc extraction procedure, MeOAc is used again, this time after the QDs are in the solid-state form. The QDs are spin-cast from octane onto a substrate, which is then immersed in MeOAc where atmospheric moisture aids the hydrolysis of MeOAc to generate acetic acid and methanol (Figure 3D).¹²¹ Acetic acid then replaces native oleate ligands to yield QD surface-bound acetate and free oleic acid. This spin-coating and ligand-exchange processing¹¹⁴ are repeated to produce films with a desired thickness (Figure 3C). The resulting films exhibit long-range electronic transport and can be used to fabricate colloidal PQD PV cells, typically using the device architecture of $\text{F:SnO}_2/\text{TiO}_2/\text{CsPbI}_3 \text{ QD}/\text{Spiro-OMeTAD}/\text{MoO}_3/\text{Al}$ (shown in Figure 3B), that operate somewhat similar to thin-film perovskite cells¹²² and achieve a large open-circuit voltage of 1.23 V and efficiency of 10.77% (Table 1).

Surface manipulation of PQDs has proven to be significantly different than chalcogenide QDs due to more dynamic surface species and low material formation energy. While acetate replaces oleate ligands as described above, cationic ligands (oleylammonium⁺) were found to remain in the QD films after the MeOAc treatment.¹²¹ In order to further improve the mobility of CsPbI_3 QD arrays, cation halide salts (AX) were used to further treat the prepared CsPbI_3 QD solid films.²⁹ AX salts could be chosen, such as formamidinium iodide (FAI), where the smaller FA^+ then replaces long-chain oleylammonium⁺ on the surface and I^- is fairly inconsequential as the QDs are I-based. As shown in Figure 3D, the AX post-treatment coats the QD array rather than alloying or inducing grain growth in the films, enabling the films to retain their nanocrystalline character. The removal of oleylammonium ligands greatly improves the electronic coupling between QDs, which doubles the film mobility, resulting in an increased photocurrent up to 14.37 mA/cm^2 and solar cell efficiency of 13.43%. While CsPbI_3 QDs clearly have multiple ligand types on the surface after synthesis,¹²¹ this fact may certainly depend on synthetic conditions and composition. For example, CsPbBr_3 has been developed without bound oleate,⁶⁸ and amine-free synthesis of CsPbBr_3 and $\text{CsPb}(\text{Br}_{1-x}\text{Cl}_x)_3$ QDs have also been reported.¹³⁵

CsPbI_3 QD PVs have demonstrated remarkably high open-circuit voltages but are certainly not entirely trap-free. The lattice defects and surface vacancies after ligand exchange in CsPbI_3 QD significantly hinder the solar cell performance; as shown in Figure 4A, a simple and efficient surface passivation method using a variety of inorganic cesium salts (e.g., cesium acetate (CsAc)) is reported to not only fill the vacancy at the CsPbI_3 perovskite surface, but also improve electronic coupling between CsPbI_3 QDs.¹³⁵ CsAc was then used in a CsPbI_3 QD solar cell which demonstrated an improved efficiency of 14.10% and exhibits improved stability against moisture.¹²⁵ The similar Cs^+ treatment was recently reported to modify the mesoporous

Table 1. Summary of Reported PQD Solar Cell Efficiencies

Absorbing Layer	V_{OC} (V)	J_{SC} (mA/cm ²)	FF	PCE (%)	Reference
CsPbI ₃	1.23	13.47	0.650	10.77	21
CsPbI ₃	1.20	14.37	0.780	13.40	29
CsPbI ₃	1.27	12.39	0.800	12.55	119
CsPbI ₃	1.18	13.59	0.726	11.64	123
CsPbI ₃	1.04	16.98	0.672	11.87	124
CsPbI ₃	1.25	14.96	0.756	14.10	125
CsPbI ₃	1.24	14.14	0.740	13.12	126
CsPbI ₃	1.18	15.21	0.742	13.30	127
CsPbI ₃	1.06	17.77	0.758	14.32	128
CsPbI ₃	1.22	15.10	0.740	13.80	129
CsPbBr ₃	1.42	7.01	0.530	5.60	130
CsPbBr ₃	1.50	5.60	0.620	5.40	120
CsPbBr ₃	0.86	8.55	0.570	4.21	131
CsPbBrI ₂	1.30	13.13	0.704	12.02	130
CsPbBr _{0.6} I _{2.4}	1.30	5.30	0.770	5.34	132
FAPbI ₃	1.10	11.83	0.644	8.38	45
FAPbI ₃	1.12	11.81	0.640	8.52	50
FAPbI ₃	1.10	15.40	0.748	12.70	51
FAPbI ₃	1.12	16.70	0.710	13.20	129
Cs _{0.75} FA _{0.25} PbI ₃	1.18	14.36	0.680	11.14	50
Cs _{0.5} FA _{0.5} PbI ₃	1.17	18.30	0.783	16.60	30
Cs _{0.25} FA _{0.75} PbI ₃ +CsPbI ₃	1.20	18.91	0.760	17.39	133
CsPbI ₃ +FAPbI ₃	1.22	17.26	0.740	15.60	134

TiO₂ (*m*-TiO₂), which enables the incorporation of CsPbI₃ QDs into *m*-TiO₂ for enhanced charge extraction and a best efficiency of 14.32% was recorded.¹²⁸ Liu and co-workers made use of μ -graphene (μ GR) to crosslink QDs to form a μ GR/CsPbI₃ QD film.¹²³ The resultant QD film provides not only an effective channel for carrier transport, as witnessed by much improved conductivity but also significantly better stability against moisture/humidity, and high temperature stresses. Xue et al. reported efficient and stable FAPbI₃ colloidal QD solar cells through rational surface regulation.⁴⁵ Tailoring the polarity of antisolvents for the post-synthetic process enabled effective removal of the insulating ligands on FAPbI₃ QDs, while preserving perovskite cores. As a result of the enhanced inter-dot electrical coupling, a power-conversion efficiency of 8.38% was achieved. Similarly, a typical MeOAc/FAI wash resulted in FAPbI₃ devices with a stabilized power output (SPO) of 8.83%.⁵⁰ A facile strategy was further developed that takes advantage of conjugated small molecules, which provide an additional driving force for effective charge separation in FAPbI₃ QD solar cells.⁵¹ The resulting PQD solar cell shows a PCE approaching 13%. The cation-exchange synthetic approach allowed for PV devices of the full compositional range of Cs_{1-x}FA_xPbI₃ to be fabricated, where the QD solar cells exhibit a lower V_{OC} loss (E_g -e V_{OC}) than the thin-film perovskite devices of similar compositions.⁵⁰ This particular study opened up more opportunities to go beyond simple CsPbI₃ or FAPbI₃ QD solar cell devices, and the latest certified record efficiency of 16.6% for QD solar cells has been achieved using Cs_{0.5}FA_{0.5}PbI₃ composition (Figure 4B).³⁰

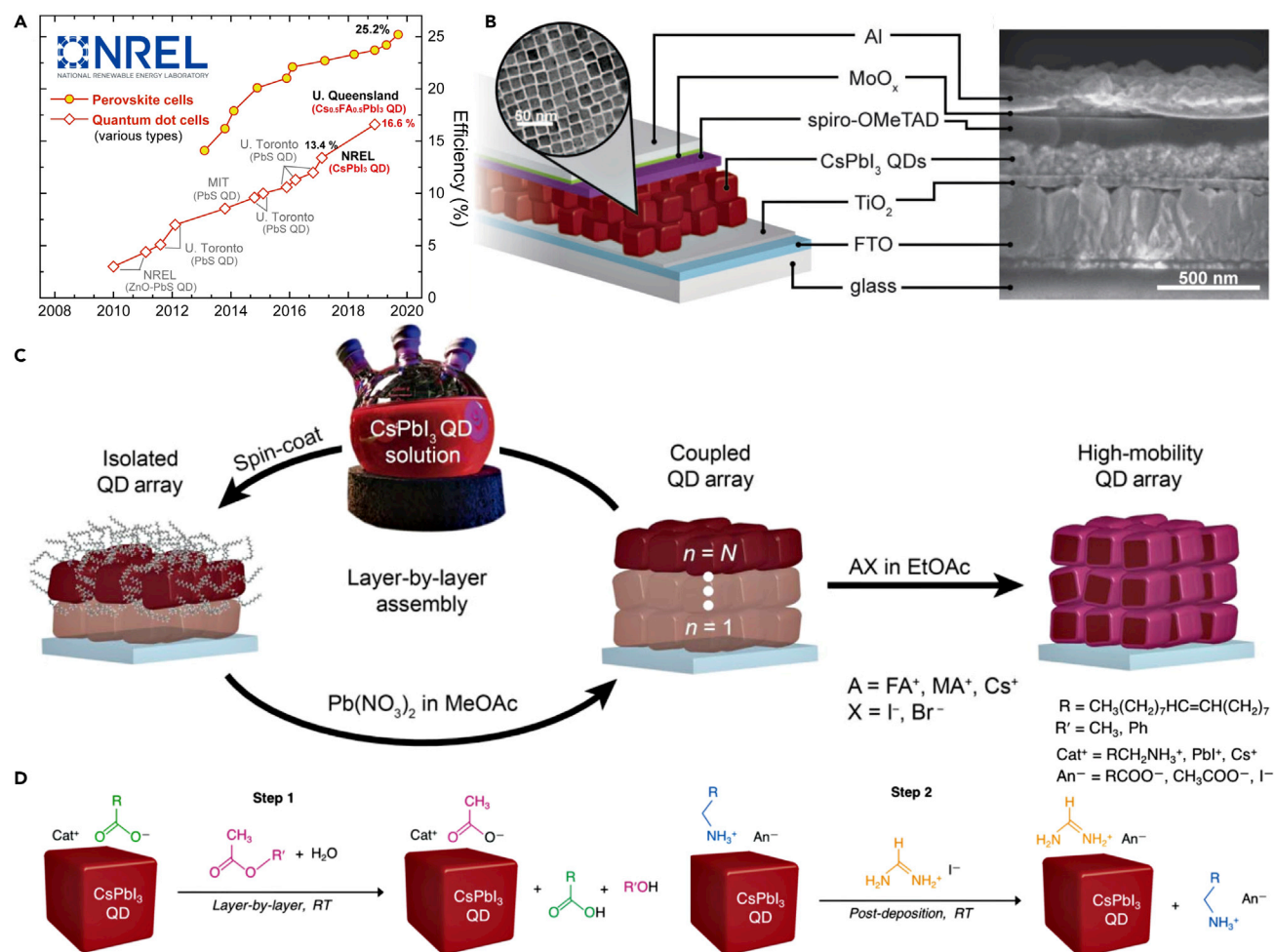


Figure 3. Development and Fabrication of CsPbI₃ PQD Solar Cells

(A) Efficiency progress of perovskite and QD based solar cells. The values of the efficiencies are adopted from NREL.

(B) Schematic and cross-section SEM of the CsPbI₃ QD solar cell, reprinted with permission from Swarnkar et al.²¹ Copyright 2016 American Association for the Advancement of Science.

(C) Schematic of the QD film deposition process with MeOAc and AX salt post-treatment, reprinted with permission from Sanehira et al.²⁹ Copyright 2017 American Association for the Advancement of Science.

(D) Anionic carboxylate and cationic ligand-exchange reaction during the layer-by-layer deposition and FAI post-treatment for CsPbI₃ QD solar cells, reprinted with permission from Wheeler et al.¹²¹ Copyright 2018 American Chemical Society.

New device structures and processing techniques can also be explored to improve the device performance. Unlike typical molecular perovskite solutions, QDs form a colloidal suspension in nonpolar solvents, which can be exploited to create structures not easily created in thin-film perovskite PVs by depositing new layers directly on each other without risk of dissolving underlying films. Zhao et al. produced PQD devices with an internal heterojunction within the QD film (Figure 4C) by depositing CsPbI₃ QD layers on top of Cs_{1-x}FA_xPbI₃ QDs layers.¹³³ The position and the composition of the heterojunction within the film was tuned by altering the spin-coating procedure and the percent of FA in the bottom film. This approach was optimized to produce films with over 17% efficiency from J-V analysis. However, non-negligible hysteresis was observed during the reverse and forward scans of the solar cell, causing the SPO efficiency to be only 15.5%. The hysteretic J-V behavior here might be mainly due to the unbalanced interfacial charge transport, and adopting electron transporting materials like SnO₂³⁰ or hole transporting materials like PTAA,¹²⁵ can

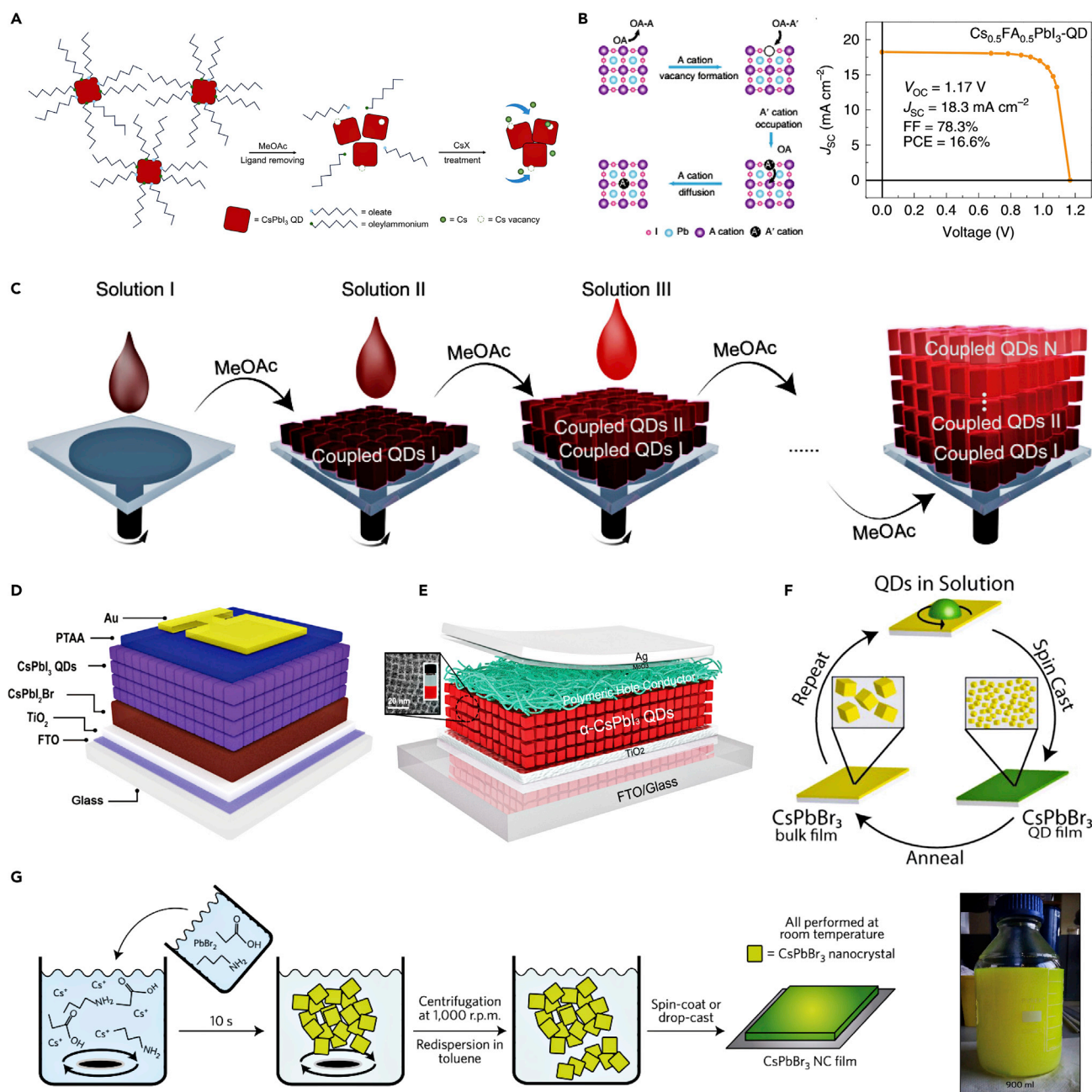


Figure 4. Device Engineering of PQD Solar Cells

(A) CsPbI₃ QD solar cell adopting Cs⁺ post-treatment, reprinted with permission from Ling et al.¹²⁵ Copyright 2019 Wiley.)

(B) Schematic of the A-site cation-exchange reaction for Cs_{1-x}FA_xPbI₃ QDs and the J-V characteristics of Cs_{0.5}FA_{0.5}PbI₃ QD solar cells, reprinted with permission from Hao et al.³⁰ Copyright 2020 Nature Publishing Group.

(C) Fabrication of graded band-gap structure PQD solar cells, reprinted with permission from Zhao et al.¹³³ Copyright 2019 Nature Publishing Group.

(D) Inorganic perovskite solar cells with CsPbI₃ QD as interfacial layer, reprinted from Bian et al.¹³⁶ Copyright 2019 Cell Press.

(E) CsPbI₃ QD solar cell adopting conjugated polymer hole conductor, reprinted from Yuan et al.¹¹⁹ Copyright 2019 Cell Press.

(F) Schematic of the perovskite thin-film deposition process for PQD solution, reprinted with permission from Hoffman et al.¹³⁰ 2017 American Chemical Society.

(G) Schematic of the scalable and environmentally friendly synthesis of PQD inks, reprinted with permission from Akkerman et al.¹²⁰ Copyright 2016 Nature Publishing Group.

significantly eliminate the hysteresis. Li et al. developed a similar CsPbI₃/FAPbI₃ PQD bilayer heterojunction structure with 15.6% efficiency and improved stability.¹³⁴ Bian et al. used this concept to demonstrate a solar cell where CsPbI₃ QDs were deposited on top of CsPbI₂Br thin films (Figure 4D), where the QDs improved the efficiency through longer lifetimes, improved carrier mobility, and well-matched energy levels.¹³⁶ Gong et al. recently used similar concept of combining PQDs with thin-film LHPs to build device that exhibits improved V_{OC} and fill factor, although similar J_{SC} as compared with thin-film LHP only devices.¹³⁷ They attribute such improvement of performance due to better light harvesting, better band alignment, and better defect passivation. This is interesting and encouraging as we have discussed the improved voltages from PQD PVs as compared to corresponding thin-film LHP devices. The combination of LHP thin films and PQD films have now led to devices over 21% using MAPbI₃ thin films and various PQDs with improved stability.¹³⁸ For wider-band-gap materials, all-inorganic CsPbBr₃ LHP thin films has shown improved performance both in terms of current and voltage when combined with FAPbX₃ PQDs as interfacial layer due to reduced charge-carrier recombination and improved charge-transfer kinetics.¹³⁹ The amount of tunability from these processing advantages will likely yield many benefits in the near future through strategic combinations of LHP films and PQDs.

Benefits of PQDs in PV

While thin-film metal halide perovskite semiconductors have gained prominence in PV research, colloidal PQD receive far less attention, but it is increasing. Understanding the full benefits of these PQDs should be further strengthened. First, as mentioned above, the CsPbI₃ QD solar cells exhibit improved V_{OC} (up to ~1.30 V) in comparison to thin-film CsPbI₃ devices (typically <1.15 V). The V_{OC} of Cs_xFA_{1-x}PbI₃ QD devices has been shown to exceed that of the state of the art for thin films in a direct comparison (same device structure, band gap, and composition).^{30,50,52} However, modifications to the device structure and further tweaks on composition have yielded thin-film V_{OC} values unmatched by QDs, namely because of the lack of research of wide-band-gap QD compositions, such as MAPbBr₃.¹⁴⁰ The V_{OC} loss ($E_g - eV_{OC}$) is an important parameter to evaluate the energy loss in solar cell devices.¹⁴¹ The V_{OC} loss is reported to be closely related to the energy disorder and defect states¹⁴² and inhomogeneous energy landscape¹⁴³ that generate nonradiative energy loss channels, and thus lead to reductions in V_{OC} .¹⁰¹ The V_{OC} loss for most inorganic perovskite devices lies between 0.7 and 0.9 eV. And for high-efficiency (>10%) PbS QDs solar cells, the V_{OC} loss is normally around 0.8 eV,¹⁰¹ depending exactly on how the band gap is measured for a particular device. For example, the V_{OC} loss is ~0.7 eV if the gap is measured at the external quantum efficiency (EQE) absorption tail. CsPbI₃ QD solar cells attain a lower V_{OC} loss, with a value of 0.52 eV²¹ from the EQE absorption tail. As shown in Figure 4E, utilizing organic polymeric hole conductor PTB7,¹¹⁹ the work function of PTB7 (−4.46 eV) matches that of CsPbI₃ QDs (−4.47 eV) resulting in a band alignment at the perovskite/polymer interface, which removes possible energetic barriers for hole extraction and thus reduces interfacial charge recombination. Therefore, using PTB7, a high V_{OC} of 1.28 V was achieved, corresponding to a lower V_{OC} loss of 0.42 eV.¹¹⁹ These high V_{OC} s achieved are in spite of the energetic disorder expected in QD films made with a range of sizes, and other effects.¹⁰¹ Further molecular engineering could strengthen the power of polymeric hole conductor strategy,¹⁴⁴ that may also be extended to conventional PbS QD solar cells toward efficient and scalable PV devices.

Meanwhile, stability issues found in current hybrid perovskite solar cells, such as current-voltage hysteresis,¹⁴⁵ photoinduced phase separation,¹⁴⁶ remain to be solved before their broad application. As discussed below, the stability of PQDs can directly address stability challenges for thin films, but to date, no rigorous long duration maximum power point tracking, high temperature or temperature cycling, or outdoor performance has been reported for PQD solar cells.^{30,147,148} Without this data, it is still presumptive to conclude that PQD solar cells are commercially more viable than any thin-film perovskite technology. Nonetheless, we can discuss the potential upside of stability in PQD solar cells. The presence of ion migration has received broad attention and was speculated to be the origin or an important contributing factor to poor stability of perovskites.¹⁴⁹ Nevertheless, no direct or indirect experimental evidence confirmed that ion migration could be eliminated or reduced in perovskite NCs. The suppressed ion migration has been witnessed in low-dimensional bulk perovskite materials and devices. Lin et al. reported the absence of notable ion migration in 2D perovskites film,¹⁵⁰ which should be attributed to the introduction of long organic *n*-butylammonium (BA) spacer cation. A similar result was also reported by Draguta et al.;¹⁴⁶ they indicated that mixed-halide perovskites can be stabilized against phase separation by deliberately engineering carrier diffusion lengths and injected carrier densities. In this regard, the zero-dimensional QDs or NCs could create a platform for further understanding and elimination of ion migration in LHP materials, and improve efficiency and stability.¹⁴⁶ In addition, the grain boundaries also have a significant impact on efficiency and stability of hybrid perovskites.¹⁵¹ When the grain size is increased from a few hundred nanometers to the micrometer level, the device performance is often significantly improved together with elongated charge-carrier lifetimes.^{152,153} Quite recently, Bakr et al. demonstrated a high efficiency of 21.09% and excellent ambient stability in single-crystal MAPbI₃ perovskite solar cells, without any grain boundaries.¹⁵⁴ In order to fabricate perovskite film with larger grains, PQDs would provide new directions without undergoing solution-to-film crystallization process. Kamat and co-workers developed a layer-by-layer methodology (Figure 4F), which makes use of CsPbBr₃ QD deposition followed by annealing, providing a convenient way to cast stable and highly crystalline CsPbBr₃ films.¹³⁰ Zeng et al. reported highly efficient inorganic mixed-halide perovskite solar cells adopting the same QD-to-bulk layer-by-layer methodology. A high V_{OC} up to 1.32 V and a best PCE of 12.02% were achieved in CsPbBr₂ inorganic solar cells.¹⁵⁵

Thin-film polycrystalline perovskites are normally processed from high boiling-point solvents (DMF, DMSO) with additives (MAI, GuaSCN, etc). To precisely control the solution-to-film crystallization process organic antisolvents (chlorobenzene) are used with thermal annealing,^{156,157} all of which further increase the complexity of the entire device fabrication. Comparatively, processing the colloidal PQD inks is easier in the sense that one need not to worry about tight processing windows for crystallization. Although the complete absorber layer building in case of PQD devices often involves more than one single deposition and processing step, the PQD ink can, however, be processed at room temperature with a variety of industrially friendly solvents like hexane, octane, toluene, etc. Further, developing QD inks geared toward mass production of PV devices is desirable. Akkerman et al.¹²⁰ have developed scalable and environmentally friendly synthesis of PQD inks consisting of a colloidal suspension of CsPbBr₃ QDs passivated with shorter ligands than previously reported synthesis (Figure 4G). The shorter ligands still suppress phase segregation of PQDs and enable room-temperature synthesis of QD inks that can be delivered at large-scale without a secondary ligand-exchange washing step. The CsPbBr₃ perovskite inks were then used to fabricate a PQD-based PV prototype

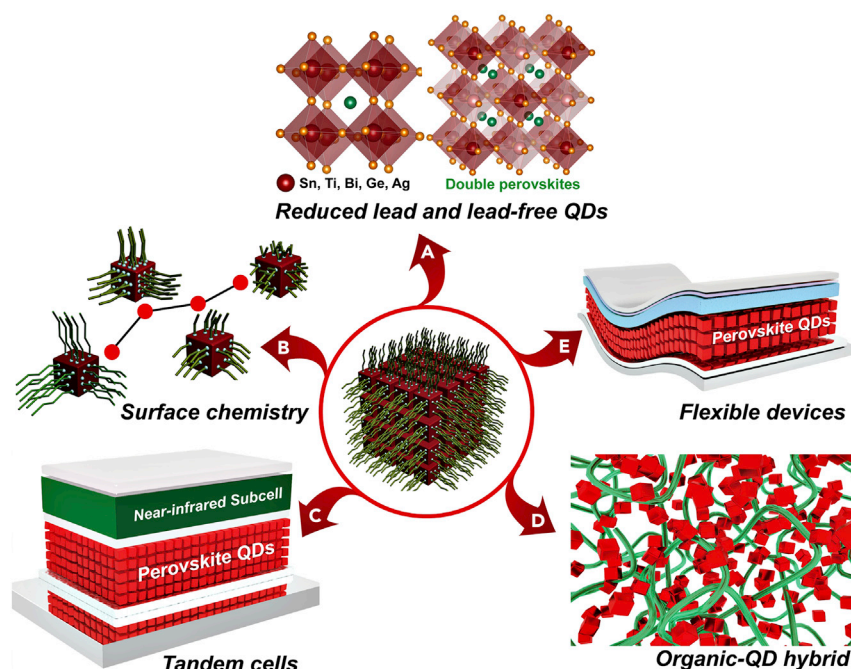


Figure 5. Challenges and Prospectives in PQD Solar Cells

- (A) The crystal structure of reduced lead and lead-free PQDs.
 (B) Schematic of surface chemistry development for PQDs.
 (C) Schematic of efficient tandem cells employing a wide band-gap PQD subcell and near-infrared subcell.
 (D) PQD/organic hybrid solar cells.
 (E) Flexible PQD solar cells.

device. With this formulation, they demonstrated a fully air-processed and electrically stable solar cell exhibiting PCEs exceeding 5% and an V_{OC} higher than 1.5 V. For the large-scale fabrication and application PV prospective, the QD ink is a really interesting concept needing further exploration.

Unsolved Challenges and Prospective in PQD Solar Cells

There has been tremendous progress in PQD research in such a short time, despite the smaller community and less attention they have received as compared to thin-film metal halide perovskites. However, more studies are needed to address many open questions to fully understand its impact to next-generation applications, such as the following:

Efficient Lead-free PQD

The presence of lead in perovskite raises obvious concerns about the toxicity of these materials upon introduction to the commercial market¹⁵⁸ and this toxicity may be one of the major factors limiting the widespread industrial applications of LHP semiconductors in the future. Exploring new metal cations for novel lead-free perovskite and controlling the micro or nano structure of existing lead-free perovskite are current two challenges limiting the further development of lead-free thin-film perovskites technologies (Figure 5A). Lead-free thin-film perovskite solar cells have experienced some progress in the past 5 years but show much less promise than standard LHP. Work on thin-film perovskite has demonstrated that Sn, Ge, Cu, Bi, and Sb ions^{159–163} could be used as alternative ions in perovskite configurations to form a new environmentally friendly lead-free perovskite structure. Quite

recently, encouraging PV efficiency over 10%^{164,165} has been reported in tin-based thin-film perovskite solar cells and 20% for Pb-Sn alloys.¹⁶⁶ The lead-free perovskites NCs/QDs as a new class of remarkable semiconductors has quickly emerged and been intensively investigated for other technological areas such as LEDs and detectors in the past few years.^{162,167–169} Potentially one can combine direct synthesis and as metal cation exchange to develop more lead-free perovskite materials. Finally, high-quality lead-free QDs will provide more chances to intensively understand the fundamental photo-physics, which could create new insights in controlling the optical, electronic, charge-transport properties and ultimately the improvement of PV performances in these emerging semiconductors.

Surface Chemistry of PQD

QD surface chemistry will need to be better understood in order to improve solar cell performance (Figure 5B). The native ligands used for size control during QD synthesis hinder the inter-QD coupling, which is not beneficial for dot-to-dot carrier transport. However, removing all of the long insulating ligands attached to the QD surface may induce degradation or phase transition of the QD, which then requires further QD surface repassivation.^{21,125} Therefore, surface ligand exchanges that use the proper ligands which balance the carrier transport and stability of the conductive QDs should be found. In PbS QD solar cells, great efforts have been dedicated to substitute the original long (e.g., oleate) ligand during the solid-state or the solution-phase ligand exchange, which have boosted the PbS QD solar cell efficiency to over 12%. Similarly, MeOAc removes some native ligands (e.g., oleate) attached to LHP QDs, but there are significant differences in the surface chemistry which is still being investigated. Exploring surface ligand chemistry is important to understand the type of charge-carrier transport in the PQD solid films. Surface chemistry along with ordered assembly of the particles mediated by tight size distribution greatly determines whether it is the band-like or the hopping-mediated transport mechanism. Achieving high carrier mobility while retaining the well-resolved discrete quantum-confined electronic states is beneficial for any QD optoelectronic device. Increasing carrier mobility and understanding the transport mechanism for chalcogenide and pnictide QD solids has been a topic of high interest with several reports demonstrating the hopping mechanism as well as the band-like transport.^{170–177} Investigating and understanding charge-transport properties in PQD solids and devices, particularly as a function of size, ligand type, and temperature is another interesting avenue, which is important for improving PV performance.

Surface chemistry also plays a crucial role in stability and quite recently, several studies have revealed that the selection of organic ligands during the synthesis is closely related to the stability of relevant PQDs.^{76,81} Understanding how the current and new ligand chemistries impact PQD solar cell operational stability is an essential next step. Organic conjugated ligands,⁸¹ molecular surface species,¹⁷⁸ inorganic short ligands (e.g., Cl^-),¹⁷⁹ or zwitterion ligands^{32,82} would not only likely enhance the charge-carrier transport from dot-to-dot, but could also improve the ambient and thermal stability of the relevant solar cells. Last but not the least, the commercialization of solution-processed LHP QD solar cells requires scalable and environmentally friendly synthesis of inks consisting of a colloidal QDs passivated with shorter ligands. This will enable the fabrication of compact, conductive, and fast-drying films of QD.¹²⁰ Solution-phase ligand-exchange method developed for PbS QD solar cells could be also explored to prepare scalable PQD inks, which could realize efficient PQD solar cells in a single and straightforward fabrication step.

Tandem Devices

While the band gap of CsPbI_3 is slightly too high for optimized PV, 1.73 eV is nearly ideal for certain multi-junction applications, particularly in conjunction with Si as a bottom cell. The optical absorption of CsPbX_3 QDs can be easily tuned for 500–730 nm by adjusting the ratio of halides (I/Br) and particle size, making it an excellent match to higher band-gap bottom cells as well (Figure 5C).^{180,181} The QD versions of these materials could be beneficial for multiple reasons including the higher V_{oc} found in QDs,¹¹⁹ phase segregation can be suppressed¹⁴⁶ and improved ambient stability of CsPbX_3 phases from QD surfaces. Additionally, solution-processed tandem cells are challenging to fabricate because similar solvents are typically used for each active layer, thus it is not trivial to prevent dissolution of the active layer below while controlling the crystallization to fabricate high-quality perovskite thin films. Conversely, QD films are prepared using mild solvents, and the as-cast QD film become insoluble in most organic solvents after ligand-exchange process, a clear advantage when fabricating multi-junction devices. Additionally, efficient CsPbX_3 QD cell can be installed with an efficient PbS QD subcell to form all-QD tandem solar cells by using information learned about the recombination layer from organic or perovskite thin-film tandem.¹⁸² When used together with organic sub-cells, especially recently developed efficient nonfullerene organic solar cells,¹⁸³ a wider variety of device structures and variable active materials can be realized. More interestingly, a hybrid tandem solar cell can be also fabricated with sub-cells consisting of low-band-gap perovskite thin-film materials to form an all-perovskite tandem cell.¹⁸⁴ Finally, perovskite solar cells are emerging as excellent top cell candidates for tandem devices utilizing silicon solar cell bottom cells.^{161,185} The use of PQDs in the hybrid QD-silicon tandem cells may allow more mild fabrication processing, leading to reduced cost and enhanced compatibility. Therefore, PQD can further provide a desired platform for realizing high-performance tandem devices with variable material combinations.

Organic-QD Hybrid Solar Cells

Perovskite colloidal QDs can be combined with other solution processable materials, such as polymers, molecules, and carbon-based materials (fullerenes, graphene)^{123,186,187} to produce composites with enhanced optical, electronic, and PV properties (Figure 5D). The use of organic materials can potentially improve the performance of CsPbX_3 solar cells. Ultrafast charge transfer between polymers and QDs has been reported in a PbS based nanocomposite¹⁸⁸ thus the use of organic materials in CsPbX_3 /organic hybrid solar cells may facilitate efficient charge-carrier separation and extraction and possibly yield higher PV efficiencies. Meanwhile, the functional group attached to the organic polymer can passivate the QD surface states,¹⁸⁹ where, for example, CsPbX_3 could be coupled with NIR conjugated molecules.¹⁹⁰ The hydrophobic organic material can repel moisture to protect the PQDs in the matrix,¹⁹¹ while the inorganic PQDs can shield organics from harsh UV illumination. In future hybrid devices, PQDs can be used as either electron donor or acceptor with a desired type-II band alignment, enabled by the energy level tunability of organic materials via modification of chemical structures.¹⁹² A vast number of excellent organic materials have been reported and used in PbS- and CdTe-based hybrid devices.^{144,189,192} In addition, organic semiconductors like conjugated polymers exhibit tunable molecular structure, NIR absorbance, ambient processability, and excellent thermal stability.^{193–196} Hence, this new PQD/organic hybrid combination may open up exciting opportunities for the realization of the full potential of PQDs materials.

Flexible Devices

Flexible and lightweight solar cells (Figure 5E) are highly promising for application in portable devices, or even spacecraft. QDs potentially offer higher mechanical flexibility than thin-film materials because of the nanoscale grain boundaries. Considering the narrowing PCE gap between inorganic CsPbI₃ perovskite thin-film and PQDs solar cells, flexible devices may be a way out for PQDs materials to display prominence in comparison with their bulk counterparts. Besides, the QD film can be deposited at low temperature using a solution-based deposition approach, which is advantageous for preparation of solar cells on flexible substrates. Conventional PbS QD solar cell has been successfully reported on flexible substrate.¹⁹⁷ Recently, Zhang et al. reported the extremely lightweight and ultra-flexible PbS QD solar cells exhibit a ~10% PCE together with a power-per-weight output of 15.2 W/g.¹⁹⁸ Quite recently, a fully automated spray-coated technology is exploited for the large-scale solution-based processing of PQD solar cells, achieving a best PCE up to 11.2% in the case of CsPbI₃ QDs.¹⁹⁹ All of these results demonstrate that solution-processed QDs are compatible with flexible substrates for the construction of lightweight solar cells with possibilities for new exciting solar energy applications.

CONCLUSION

PQDs have demonstrated remarkable performance in various areas of optoelectronics. They have enticed materials scientists, chemists, and physicists to develop this material system into excellent performing PV devices. Within the next-generation solar cell category the certified efficiency for QD solar cell now compares favorably to organic solar cells and other thin-film technologies due to recent success. In our opinion, the most important task currently for single-junction PQD solar cells is to continue to press forward to higher efficiencies over 20% with a band gap of >1.7 eV. QD technology offers improved operational advantages over solution-processed bulk materials solar cells (e.g., organic, bulk perovskite). With the efficiency goal achieved, we believe LHP QDs will gain increasing scientific interest and, very likely, will turn into a great commercial opportunity in the near future for optoelectronics industry.

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AUTHOR CONTRIBUTIONS

J.Y. initiated this review; drafted, edited, and revised the manuscript; and generated all Figures. A.H. assisted in the preparation of the synthetic part, provided input for

discussions, literature review. Q.Z. assisted in the editing and formatting of the manuscript, provided input for discussions. X.L. and T.M. assisted in the literature review, assisted in editing the manuscript, and provided input for discussions of the devices part. W.M. and J.M.L. supervised the preparation this review; drafted, edited, and revised the manuscript.

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